Ricardo Boeri Decal

San Francisco, CA, USA · [redacted] · public@ricardodecal.com
crypdick · crypdick · rdecal · GScholar · RicardoDecal.com

Profile

Senior ML researcher with 10 years across research and industry, including molecular biology, computational neuroscience, and production ML. Track record owning end-to-end ML work: data engines for long-tailed open-world inference, distributed training infrastructure on Ray, and ML reliability under real-world distribution shift. Currently a Machine Learning Researcher at Latent Health, training AI models to modernize healthcare.

Machine Learning Experience

Machine Learning Researcher — San Francisco, CA
Latent Health — Jun. 2026 — Now

Training AI models to modernize healthcare.

Machine Learning Engineer — San Francisco, CA
Anyscale — Jan. 2025 — Jun. 2026

Anyscale builds Ray, the open-source distributed compute framework underlying RL post-training, large-scale training, and inference at frontier labs.

- Conceived, pitched, and leading **Ray-bench**: a benchmark for evaluating LLM agents on distributed-systems code (framework translation, Ray code generation from spec, bug fixing, code optimization), paired with **RayGym**, a sandboxed multi-node execution harness for code-RL post-training. Strategic goal: drive Ray adoption among coding agents by giving model providers a verifiable-rewards benchmark to train on. Currently leading data generation: scraping, LLM-assisted synthesis with rejection sampling, AST-based Ray-API verification, and decontamination against pre-training corpora.
- Conceived, pitched, and designing a separate RL environment to train a Ray Core resource auto-calibration model.
- Engineered scalable distributed Ray implementations of: GRPO-based RL post-training for LLMs, LLM-as-judge data curation, 670B-parameter LLM serving, multimodal training, and agentic workflows.
- Early user and feedback partner for Ray LLM and Ray Direct Transport (RDMA-based tensor transfer for disaggregated inference/training); surfaced bugs and informed design.
- Built agentic systems for automated debugging and codebase-wide standards enforcement.
- Designed benchmarks comparing performance and cost across Ray OSS and RayTurbo.

Lead Machine Learning Engineer — Remote
Dendra Systems — Feb. 2020 — Jan. 2025

Founding ML lead. Built end-to-end ML systems for large-scale, long-tailed visual recognition across thousands of classes — data engine, training infrastructure, evaluation, and serving. Domain: aerial imagery for ecosystem monitoring.

- **Data engine for long-tailed, open-world inference across thousands of target classes.**
  - 60% training-data reduction at no performance cost via a novelty-maximizing data-pruning method I developed (original research, unpublished); yielded pareto-optimal (exponential) scaling laws.
  - Active learning to systematically harvest high-leverage training data, preventing model hallucinations on out-of-distribution inputs.
  - Few-shot models paired with an active-learning data-harvesting UI to onboard new classes and domains in hours instead of months (`Gradio`).
  - Unsupervised "trip-wires" for detecting model hallucinations in production; alerts routed to annotation prioritization workflow (`Jira`).
  - Annotation QA/QC: systematically surfaced mislabeled and partially labeled samples to create a "self-healing" training dataset.
- **Training and inference infrastructure (Ray-native, cloud-scale).**
  - Distributed, scale-agnostic infrastructure for training, hyperparameter tuning, and inference (`Ray/Anyscale, AWS Batch`).
  - 20× training-cost reduction via Bayesian hyperparameter tournaments with aggressive early stopping (`Ray Tune, HyperOpt, ASHA`).
  - Profiled and optimized multi-GPU throughput (`Grafana`).
  - End-to-end automation of training/inference pipelines (`custom orchestrator`); later re-implemented serverless for reliability and cost (`Step Functions, API Gateway,` λ`, EventBridge`).
  - Researched, experimented, and productionized novel ML techniques: models, samplers, optimization functions, augmentations.
- **MLOps and reliability.**
  - Sanity checks for detecting silent failures during training; stratified validation across data slices.
  - Debugged model failures using interpretability methods (GradCAM, transformer attention maps) and visualization of training dynamics.
  - Full artifact lineage and reproducibility (`MetaFlow`).
  - Observability across pipelines (`Cloudwatch, Slack bots, UMAP, Sentry`); weekly metric reviews to prevent customer-impacting incidents.
  - 80× acceleration in insights delivery via model-in-the-loop annotation tooling.
- **ML leadership.** Founding ML lead; drove transformation from services to ML-product company. Owned ML roadmap, KPIs, and OKRs aligned with product and operations. ML automation increased throughput and profitability and unlocked new markets. Oversaw data-collection team and ran internal "ML University" lectures.

Lead Data Scientist — Remote
PaceMate — Jan. 2019 — Dec. 2019

PaceMate monitors transmissions from bluetooth-enabled heart implants, identifying life-threatening arrhythmias and alerting emergency services. Solo ML hire; took cardiac arrhythmia detection from zero to deployed deep-learning system in a regulated, life-critical pipeline — end-to-end ownership across data, modeling, clinician tooling, and SOC2 security.

- Trained SOTA deep neural network for arrhythmia classification, tuned to the implant device used by the majority of patients (`Keras`).
- Built ECG data-processing pipelines for highly imbalanced clinical data (`imbalanced-learn`, custom tools).
- Built clinician-facing labeling and prediction-review dashboard for electrophysiologists (`Plotly Dash`).
- Built EMR analytics dashboard with performant ETL across heterogeneous medical records to enable data-driven decision-making (`SQL`, `PySpark`, `spaCy`, `Plotly Dash`).
- Translated clinical requirements into ML specs with cardiologists; presented to C-suite and met with investors.

Data Scientist Sarasota, FL
New College of FL, F.A.R. Institute Aug. 2018 — Dec. 2018

Master's capstone (unpaid). Predicted 30-day hospital readmission via patient-visit embeddings (`visit2vec`, t-SNE); modeled chronic-condition trajectories of heart-failure patients with clustering, network graphs, and finite-state models (`SQL`, `PySpark`, `NetworkX`, `TensorFlow`).

Research Intern Seattle, WA
Peng Lab, Allen Institute for Brain Science June 2018 — Aug. 2018

Built a 3D Deep Q-Network for tracing neural structures in petabyte-scale fluorescent microscopy; designed the simulation environment and reward system end-to-end based on manually traced microscope images (`TensorFlow`, `rl-medical`, `OpenAI Gym`; *Profundo* writeup).

Research Assistant Seattle, WA
Fairhall Lab, University of Washington Oct. 2014 — Jan. 2016

Computational neuroscience lab. Self-funded via NIH PA-12-149 federal grant.

- Parameter inference: fit biophysical models of mosquito thermonavigation to wind-tunnel behavioral data via numerical optimization (`scipy.optimize`, `sklearn`, `statsmodels`).
- Wrote vectorized `numpy` routines for high-throughput processing of behavioral video data; computed kinematic and thermal-sensing statistics across the dataset.

Molecular Biologist Various
Various Pre-2014

Before transitioning to AI/ML, I was a molecular biology researcher. Co-authored peer-reviewed work on Parkinson's disease (mitochondrial dysfunction in dopaminergic neurons; *PNAS* 2012) and Wnt signaling (*Genetics* 2012). Honors thesis on RNA interference mechanisms in *C. elegans*.

Education

M.S. Data Science Sarasota, FL
New College of Florida Aug. 2017 — Dec. 2018

B.A., Chemistry/Biology (with honors) Sarasota, FL
New College of Florida Aug. 2007 — May 2011

Early admission (admitted 16 yrs old) Jupiter, FL
Harriet L. Wilkes Honors College Jul. 2006 — May 2007

Publications, Presentations, & Teaching

* 2025 *Deploy DeepSeek-R1 with vLLM and Ray Serve on Kubernetes* — end-to-end deployment guide for distributed serving of a 670B-parameter reasoning model.
* 2025 *Ray Direct Transport: RDMA Support in Ray Core* — technical write-up on RDMA-based tensor transfer for disaggregated inference/training.
* ongoing Personal blog (80+ technical notes on Ray, LLMs, RL, ML internals, OSS projects). Selected: *Self-authoring LLM knowledge base*, *Using DSPy with Ray Data*, *Slouchless: VLM for posture correction*.
* 2025 Speaker at Ray meetups: Daft+Ray, vLLM+Ray, Skypilot+Ray.
* 2024 Talk at *AI In Production Conference* on data curation.
* 2021 Invited talk at *Ray Summit*: *How Ray and Anyscale Make it Easy to do Massive-scale ML on Aerial Imagery* (accompanying blog).
* 2019 Seminar at *New College of FL*: *Remote Sensing of Cardiac Arrhythmia at Scale using Deep Learning*.
* 2019 Seminar at *Escuela Secundaria Tecnica de Torquinst*: *Intelligencia Artificial*.
* 2015 UW Outreach: various educational events for students from low socioeconomic backgrounds.
* 2012 Two peer-reviewed journal articles in *Genetics* and *PNAS*, also presented as posters at three national conferences.
* 2007 Undergraduate honors thesis on RNA interference mechanisms in *C. elegans*.